DSAI4901 · CAPSTONE PROGRESS BRIEFING

From generating to verifying

A reference-free measure of deck usability, used as a control signal in paper-to-slide generation

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WHERE THE FIELD STANDS

Two families, one missing instrument

Work	What it provides	What it leaves open
PPTAgent, SlideGen, SlideLab, ArcDeck	Strong generators; narrative planning	Whether any given deck is usable; local, mechanical failures
ReDeck, PPTPilot, SLIDEFORGE	Repair loops that read rendered feedback	Which class of defect responded to repair, and by how much
SlideAudit, DECKBench, PresentBench	Flaw taxonomy; deck-level scoring against a reference	Arrives after generation; needs a target deck or labels

Read together, the field is good at producing decks and good at scoring them afterwards, but it has not closed the loop in between.

The instrument arrives too late to steer generation, and it is not decomposable when it does arrive.

> The gap: no instrument runs inside the loop, and none of them decomposes the score. A deck-level number cannot tell you which defect to fix.

WHY THIS DIRECTION

Three things the literature does not do

Attribution is missing

ReDeck and PPTPilot both repair, but a deck-level gain of ten points does not say which defect class responded.

Measurement is post-hoc

SlideAudit and DECKBench score finished decks. A score that arrives after generation cannot steer it.

Instruments need references

A target deck, or human labels on images. Neither can run on an arbitrary .pptx at generation time.

So the contribution is not another generator, and not a new taxonomy.

- **It is a measurement layer** that is reference-free, computed on the object model, and cheap enough to run inside the loop.
- **And a study of what that measurement is worth** - per violation class, against cost.

THE QUESTION

One gap, two research questions

> *“Slide quality can now be measured after the fact. What these instruments cannot answer is whether measurement can drive generation.”*

RQ1 Can a small set of mechanical gates replace a reference deck or human labels, and still rank decks in the order a person would?

RQ2 What does using that signal inside the loop actually add - per violation class, and at what cost?

Both are answered by the same artefact: a deck, a score, and a repair log.

THE SYSTEM

Seven parts, four of which call a model

A generation - text layer, frozen once written

parse	claims AI	requirement AI	select	outline AI	bullets+notes AI	render
PDF -> sections, figures, captions	assertions with an evidence id	talk length and audience -> table	keep or drop, per claim	re-order into an argument	compress, then write the talk	native, editable .pptx

B measurement - six gates on the .pptx object model, no rendering

G1 containment	G2 overlap	G3 font size	G4 contrast	G5 text fit	G6 density	SRR	ready?

1 measure
worst violation first

2 repair
cheapest reversible action

3 re-measure
gain per second and per dollar

Only the text layer is allowed to change the words. Repair may move, resize or recolour; it may never add content or change a number. That is what keeps “the loop improved the deck” from being confounded with “the loop rewrote the deck”.

Captions are re-rendered, not extracted

- **Headings** are found by font size and weight, not by a fixed list.
- **Figures** are vector art in this literature: pulling embedded images returns 32x32 icons. So each caption's region is clipped and rasterised at 200 dpi.
- **The band stops at the nearest paragraph**, not at any text block - a figure carries its own labels, and stopping at those cut the figure in half.

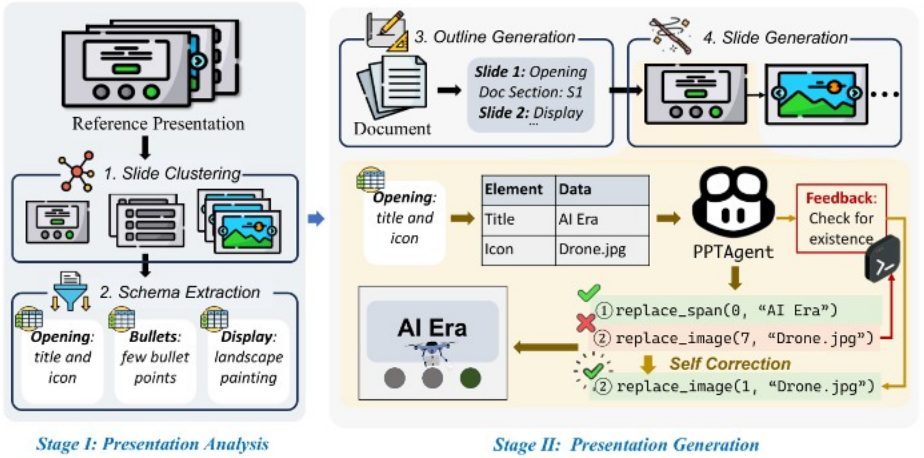


Figure 2 of PPTAgent, clipped from the PDF

12 of 32 captions in this paper produced a usable crop. The failures are tables whose caption sits above the table, and figures that span both columns. Both are fixable; the rule that produced the 12 is already the one the evaluation will use.

Selection is a table, not a judgement

request	always kept	kept if room	dropped
5 min, general	results, conclusion	method (core only)	analysis, limitations
12 min, expert	results, conclusion, method, motivation, background, analysis	related work, setup	-

Every kept or dropped claim carries the id of the rule that decided it:

claim	section	imp.	decision	rule id	why
c3	results	3	kept	hard:results	results are never dropped
c7	setup	2	dropped	table:12:setup	not kept in a 12-minute talk
c9	analysis	1	dropped	importance:cut2	importance 1 < cutoff 2

The model appears at two boundaries only: it converts a free-text request into that table, and it extracts claims. The keep/drop decision in between is a lookup, so a second person can re-run it.

PART 3 · MEASURING

Four of six gates exist; two are next

gate	rule	threshold	state
G1 containment	every shape inside the slide	0 pt	**built**
G2 overlap	text intersection over the smaller box	> 5 %	**built**
G3 font size	body and title, by slide role	role min	**built (global)**
G6 density	characters and bullets, by role	role max	**built (global)**
G4 contrast	WCAG 2.2 AA, fill against background	4.5 : 1	**next**
G5 text fit	rendered height over box height	<= 0.95	**next**

SRR

SRR = Σ P / Σ J

J = checks that could be judged, P = those that passed. One weight per check; none is fitted.

unknown is not n/a

n/a: does not apply - one text box, so no G2 pair

unknown: applies but cannot be computed - image background

both leave the denominator, both carry a reason code

deck-ready = no failure anywhere, and every gate judged at least once. A title slide does not need G6; the deck does.

SCOPE

Sixty-five things you could measure. Six are worth it

candidate	n
A layout geometry	14
B typography	15
C colour and contrast	7
D content and meaning	12
E structure	8
F rendering	5
G audience	4
total	65

Five tests

- decidable without a reference
- deterministic: same file, same number
- a violation is worse, not different
- cheap enough to run in the loop
- explainable: shape, value, threshold

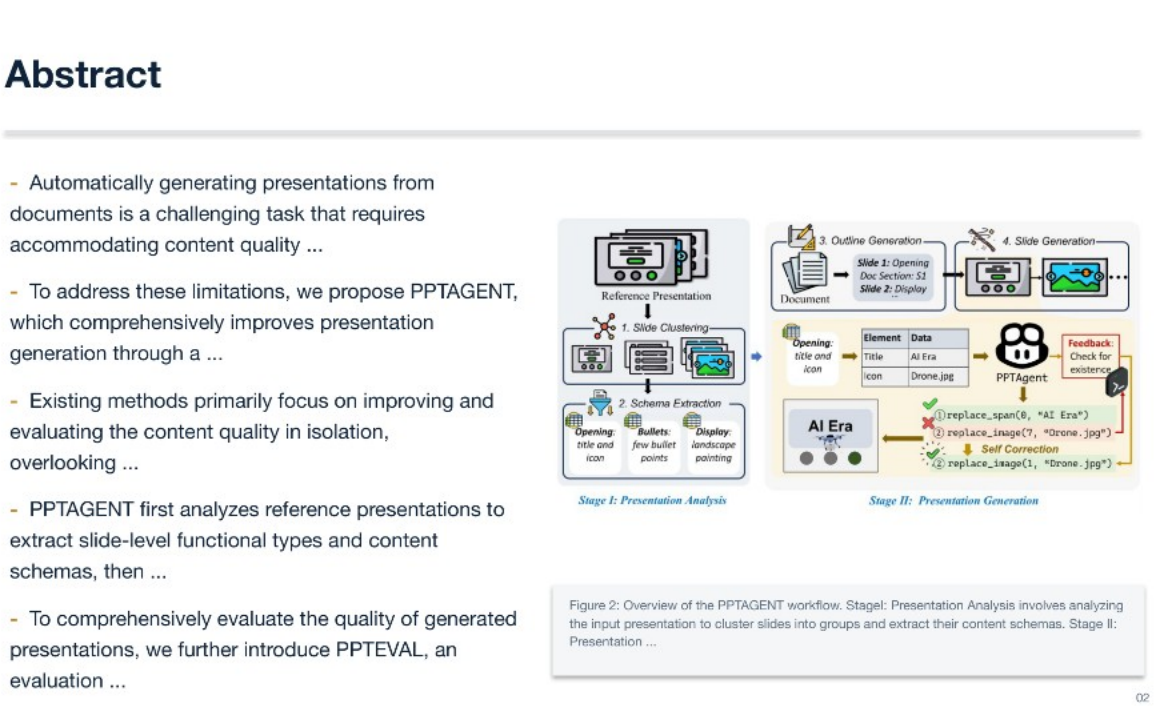
What survives

gate	from
G1 containment	A1
G2 overlap	A2
G3 font size	B1
G4 contrast	C1
G5 text fit	B7
G6 density	B5+B6

> And what it cannot see: a bullet can be grammatical, the right length, the right size - and still nonsense. That is the boundary of a mechanical metric, and it belongs to the human study.

RESULTS SO FAR

Every generated slide fails at least one gate



Slide 2 of the PPTAgent deck, generated by this pipeline

	PPTAgent	DOC2PPT
slides	9	12
clean slides	0	0
SRR (slide-level)	0.00	0.00
violations	33	40
slides failing G3	9 / 9	12 / 12
slides failing G6	8 / 9	11 / 12

This is the expected result, not a defect: the baseline generator ignores every gate. What matters is that the number is reproducible and the violations are named.

THE RUN

It runs end to end, and it reports 40 real problems

```
capstone-ppt-agent — zsh
%  ./run.sh research/pdfs/DOC2PPT_2101.11796.pdf
[1/3] parse
pymupdf=2.0 sections=12 captions=25 figures=13 (clipped=13) -> build/DOC2PPT_2101.11796.deck.json
  p1   Abstract                                                  952 chars
  p1   Introduction                                             5250 chars
   ...
  p7   Conclusion                                               450 chars
[2/3] compose
slides=11 -> build/DOC2PPT_2101.11796.plan.json
    Abstract                                        bullets=6 65s
[3/3] render
slides=12 -> out/DOC2PPT_2101.11796.pptx
done -> out/DOC2PPT_2101.11796.pptx

%  python3 src/detect.py out/DOC2PPT_2101.11796.pptx
out/DOC2PPT_2101.11796.pptx
  slides=12  clean=0  SRR=0.0
  violations=40  by gate={'G3': 12, 'G6': 11}
    p1<G3> TextBox 4 at 14 pt
    p2<G3> TextBox 3 at 15 pt
    p2<G6> 783 chars
    ...

%  python3 -m pytest
.....................................................................  [100%]
76 passed, 7 warnings in 4.78s
```

Real terminal output; rows in the middle are elided.

parse
12 sections, 25 captions, 13 figures clipped

compose
11 slides at 65 seconds

render
a 12-slide deck, native and editable

detect
40 violations, SRR 0.00

pytest
76 cases green

STATUS

What is built, what is next

Built and tested

- parse: sections, figures, captions
- figure clipping, 25 captions -> 13 crops
- render: native .pptx with speaker notes
- G1, G2, G3, G6 + SRR
- selection rule table, 55 test functions

Partially built

- G3 / G6 still use one global threshold
- figure captions need a word-boundary trim
- claims are still extracted by heuristic

Not started

- G4 contrast, G5 text fit
- repair actions and the loop
- 110-slide test set and its labels
- human decks, baselines, the 2x2 experiment

76 test cases pass. The deck above was produced by this pipeline end to end.

PLAN

The experiment, and the calendar

The design: a 2 x 2 factor

	no inspection	inspection on
base model	A1	A2
fine-tuned	A1'	A2'

A2 - A1 isolates what inspection adds. The fine-tuned row tests whether that gain survives a better generator.

The fine-tuned arm is conditional: it runs only if the main result is in and four weeks of buffer remain.

weeks	work	milestone
W1-2	scope and literature	proposal, 5 Nov
W3-7	rulebook, six gates, calibration	validity check
W8-11	repair actions, 110-slide test set	interim report, 7 Jan
W12-18	loop, ablation, 20-person study	preference correlation
W19-23	report, demo, presentation	final, 8 Apr

Two validity checks run before the study: decks must rank human > PPTAgent > naive, and SRR must track preference. The correlation is pre-registered and reported whatever it shows.

WHAT I NEED FROM YOU

Three decisions and one question

1

Human decks

Where can I get decks an author actually presented? They are the upper bound in the validity check, and I have none.

2

Scope

Single-column English papers, 10-12 slides, 8 held-out papers - is that narrow enough to finish, or too narrow to generalise?

3

Gate set

Six mechanical gates, with density and font size swept rather than fitted. Is that defensible, or should one be dropped?

And one question I would like your view on: is a deterministic detector seen as a strength here, or as a gap in the AI content of the project?